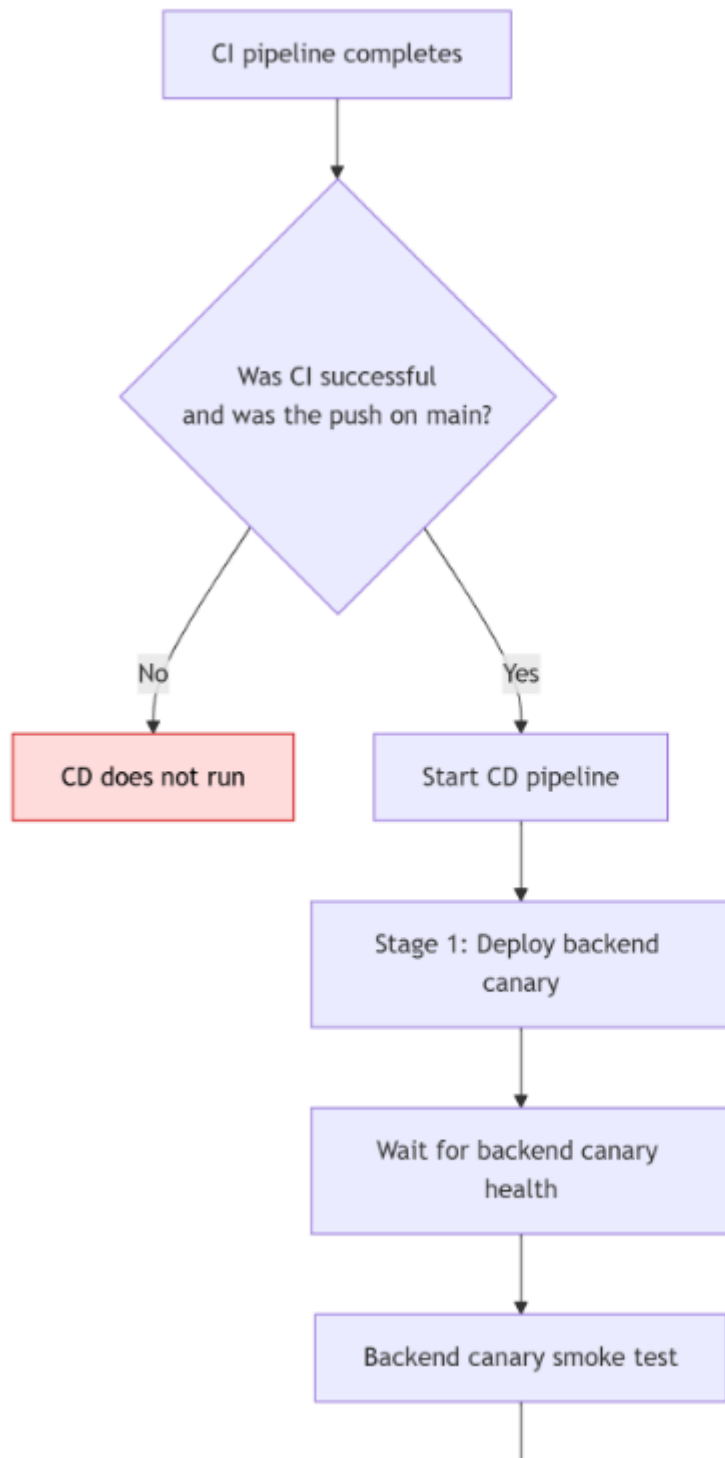
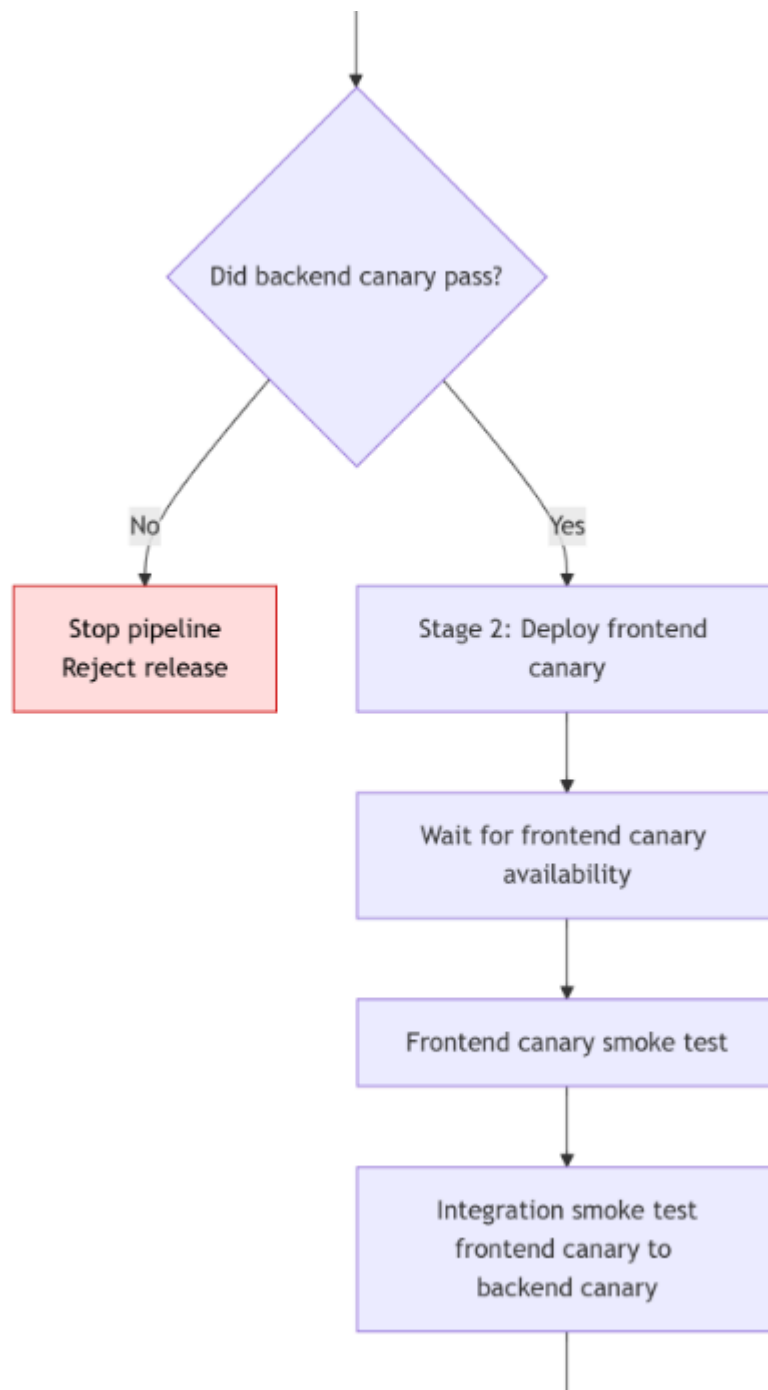
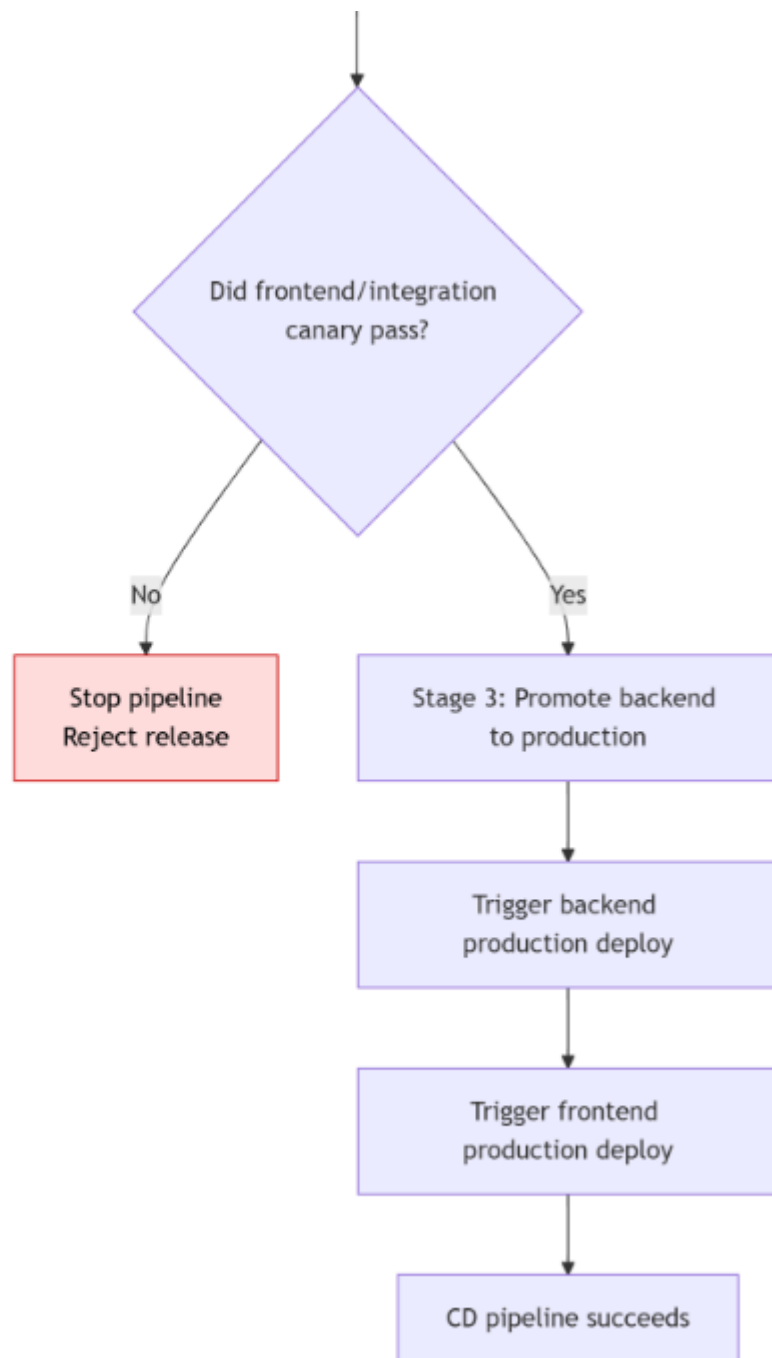


CD pipeline diagram:







CD pipeline:

<https://github.com/UTSC-CSCC01-Software-Engineering-I/term-group-project-c01w26-project-ctrl-alt-react/blob/main/.github/workflows/cd.yml>

Tech overview: We used Render to host the frontend and backend of both the staging (canary) environment and production environment, with images drawn from docker registry pushed in the CI step.

Canary analysis: After CI passes, we deploy backend and frontend canary versions, then run health, backend smoke, frontend smoke, and integration/CORS checks. We promote only if all checks pass.

Canary promotion strategy: If canary validation succeeds, we trigger backend and frontend production deploy hooks. This is a gated rollout, so failed canary checks block promotion. Downtime is typically near zero on Render because the current production version serves traffic while the new version starts.